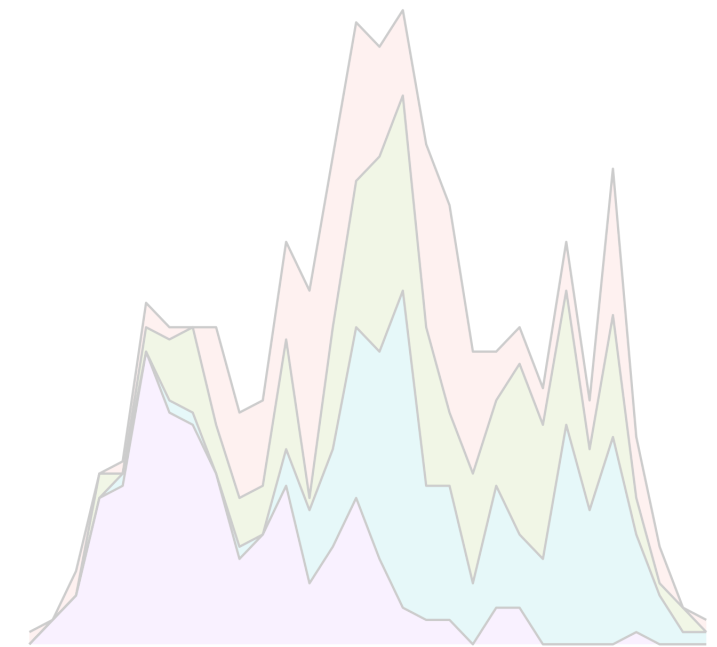
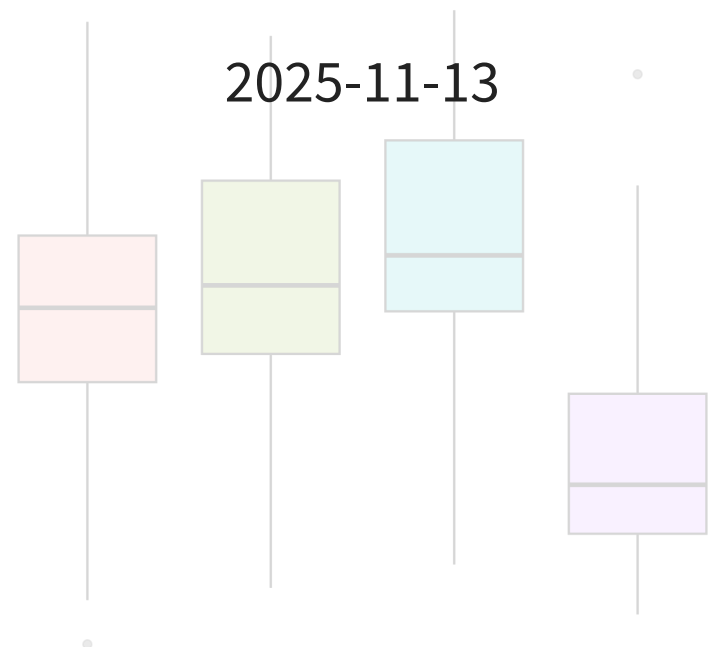
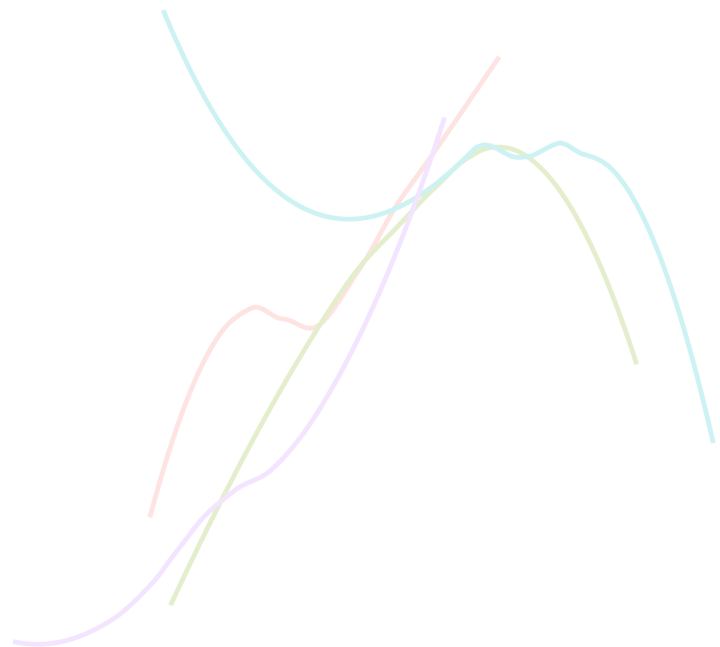


Inference in SLR



Housekeeping

- Midterm 2 is one week from today (in class)
 - Practice problems to be released over the weekend

Recap

- Learned how to interpret slope and intercept of fitted model
 - b_0 is estimate of $\hat{y}$ when $x = 0$
 - b_1 is expected change in $\hat{y}$ for a one unit increase in x
- When explanatory x is categorical, we have a slightly more nuanced interpretation
 - Created indicator variable with 0 = base level and 1 = non-base level
- Coefficient of determination R^2 assesses strength of linear model fit

Live code!

How to fit a sample linear regression model in [R](#). (Code posted after class!)

Inference for SLR

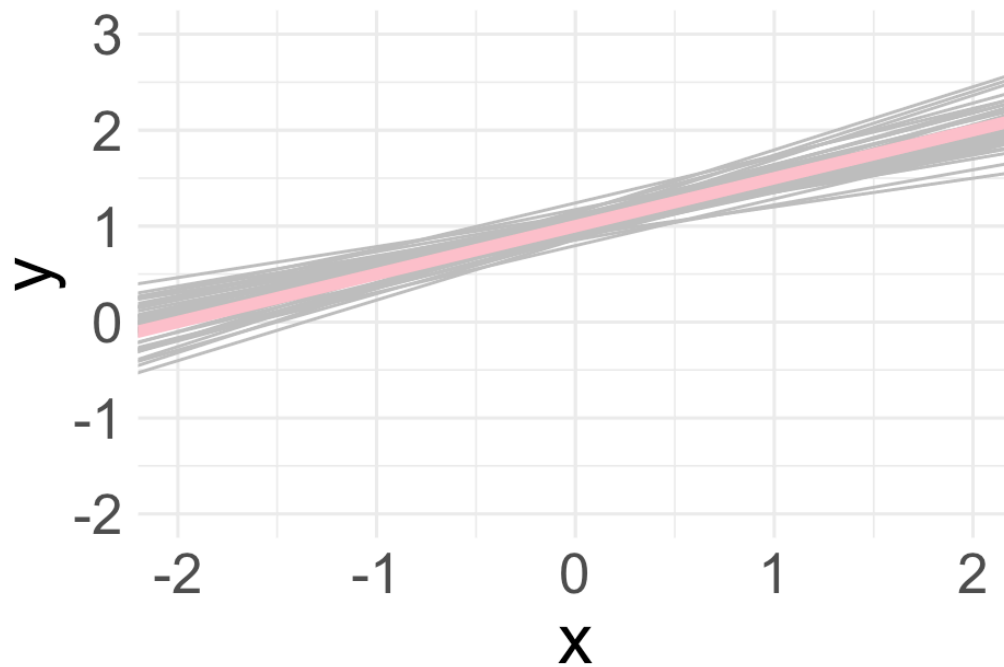
Variability of coefficient estimates

No need to take notes on this slide!

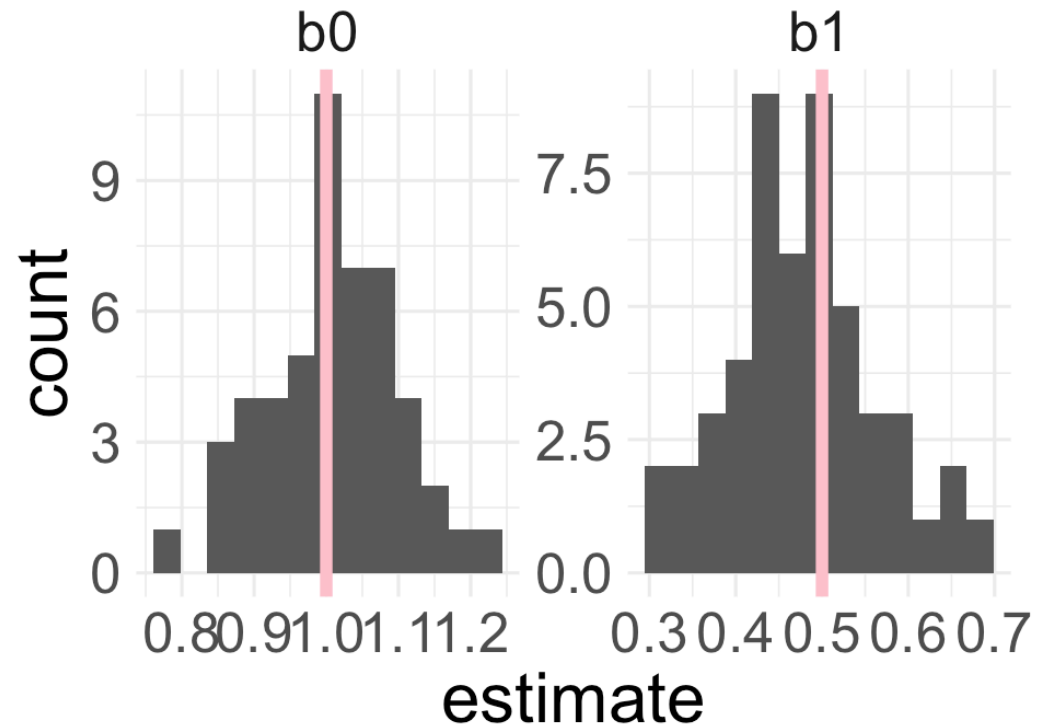
- Remember, a linear regression is fit using a sample of data
- Different samples from the same population will yield *different* point estimates of (b_0, b_1)
 - I will generate 30 data points under the following model: $y = 1 + 0.5x + \epsilon$
 - How? Randomly generate some x and ϵ values and then plug into model to get corresponding y
 - Fit SLR to these (x, y) data, and obtain estimates (b_0, b_1)
 - Repeat this 50 times

Variability of coefficient estimates

Fitted/estimated lines
From 50 simulations



Sampling distribution



Pink = line using true intercept and slope

What are we interested in?

Remember: we fit SLR to understand how x is (linearly) related to y :

$$y = \beta_0 + \beta_1 x + \epsilon$$

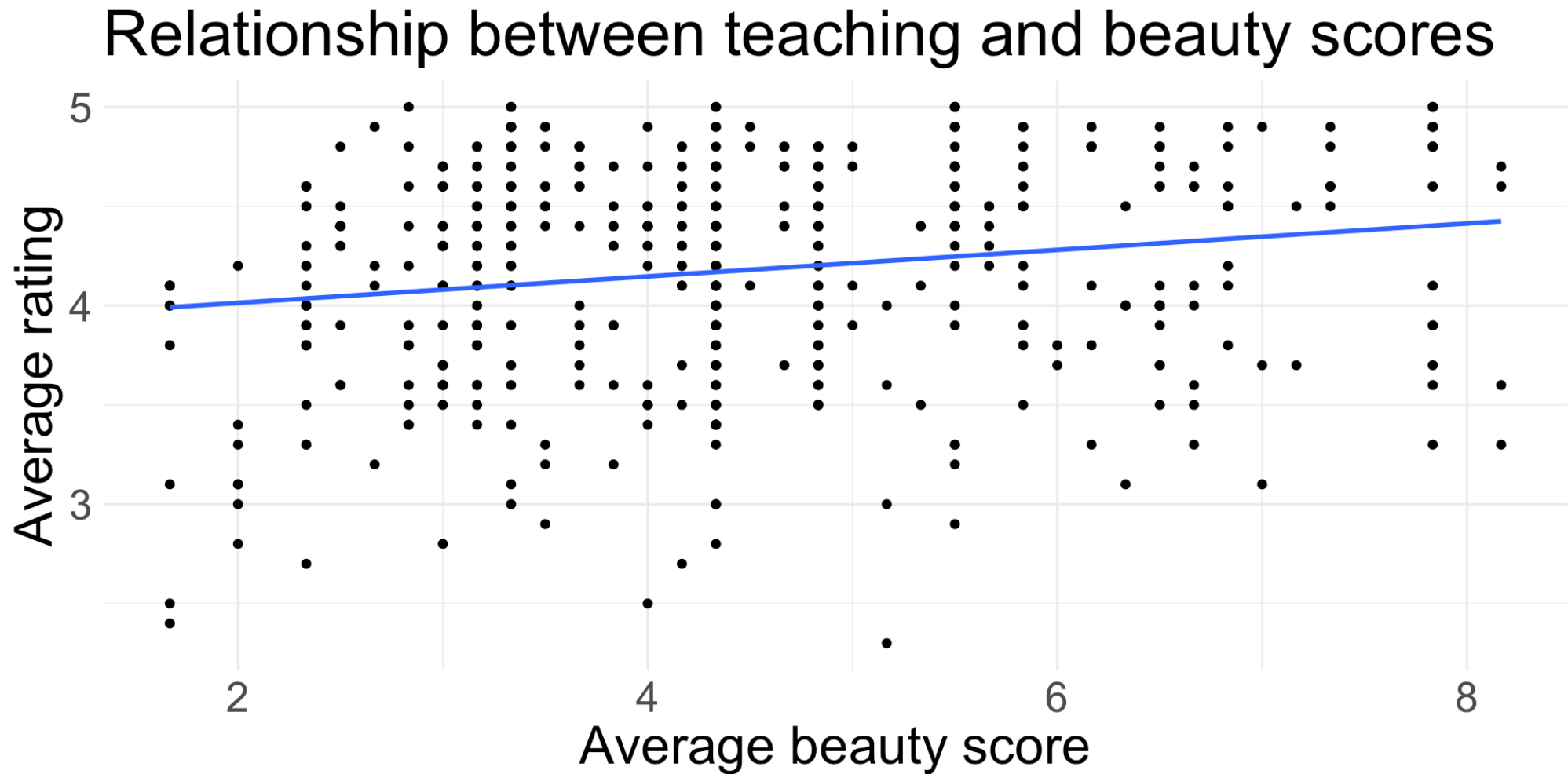
- What would a value of $\beta_1 = 0$ mean?
 - If $\beta_1 = 0$, then the effect of x disappears and there is in fact no linear relationship between x and y
- We don't know β_1 , so we can perform inference for it!
 - Can conduct HTs and obtain CIs using our best guess b_1

Running example: **evals** data

Data on 463 courses at UT Austin were obtained to answer the question: “What factors explain differences in instructor teaching evaluation scores?”

- One hypothesis is that attractiveness of a teacher influences their teaching evaluations
- We will look at the variables:
 - **score**: course instructor’s average teaching score, where average is calculated from all students in that course. Scores ranged from 1-5, with 1 being lowest.
 - **bty_avg**: course instructor’s average “beauty” score, where average is calculated from six student evaluations of “beauty”. Scores ranged from 1-10, with 1 being lowest.
- Write out our linear regression model

Teaching evaluations data



Does this line really have a non-zero slope?

Hypothesis test for slope

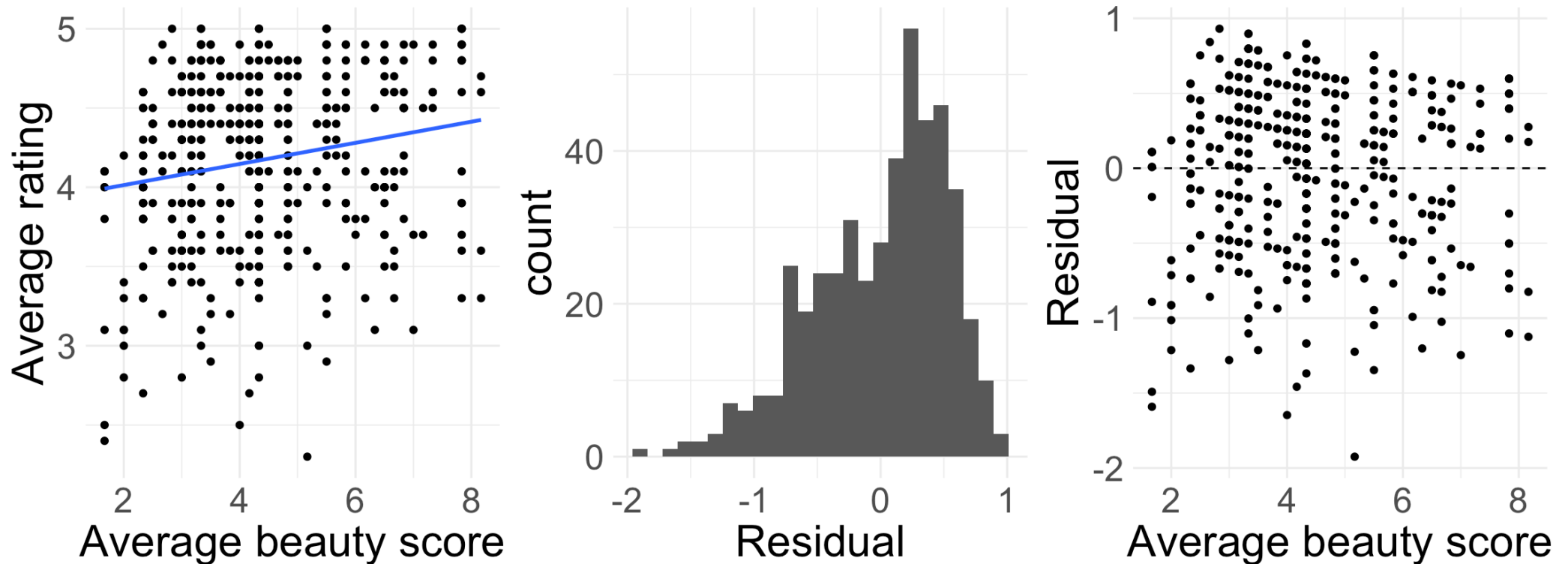
- $H_0 : \beta_1 = 0$: the true linear model has slope zero.
 - In context: there is no linear relationship between an instructor's average beauty score and their average teaching evaluation score.
- $H_A : \beta_1 \neq 0$: the true linear model has a non-zero slope.
 - In context: there is a linear relationship between an average instructor's beauty score and average teaching evaluation score.
- To assess, we do what we usually do:
 1. Check if methods are appropriate
 2. If so: obtain an estimate, identify/estimate standard error of the estimate, find an appropriate test statistic, and calculate p-value
- The output from `lm()` actually does all of #2 for us, but we will see how the test statistic and p-value are calculated!

Teaching evaluations: model assessment

We fit the model in R, and obtain the following plots.

Are all conditions of LINE met?

Relationship between teaching and beauty scores



Looking at `lm()` output

```
1 library(broom)
2 eval_mod <- lm(score ~ bty_avg, data = evals)
3 eval_mod |>
4   tidy()
```

term	estimate	std.error	statistic	p.value
(Intercept)	3.880	0.076	50.961	0.00000
bty_avg	0.067	0.016	4.090	0.00005

Assuming the linear model is appropriate, interpret the coefficients!

- Intercept: an instructor with an average beauty score of 0 has an estimated evaluation score of 3.88
- Slope: for every one point increase in average beauty score an instructor receives, their evaluation score is estimated to increase by 0.067 points

Looking at `lm()` output

term	estimate	std.error	statistic	p.value
(Intercept)	3.880	0.076	50.961	0.00000
bty_avg	0.067	0.016	4.090	0.00005

- **estimate**: point estimate (b_0 or b_1)
 - **std.error**: estimated standard error of the estimate ($\widehat{SE}$)
 - **statistic**: value of the test statistic (follows $t_{df=n-2}$ distribution for SLR)
 - **p.value**: p-value associated with the *two-sided alternative* $H_A : \beta_1 \neq 0$
- Let's confirm the test statistic calculation:

$$t = \frac{\text{observed} - \text{null}}{\widehat{SE}_0} = \frac{b_{1,obs} - \beta_{1,0}}{\widehat{SE}_0} = \frac{0.067 - 0}{0.016} = 4.09$$

p-value and conclusion

term	estimate	std.error	statistic	p.value
(Intercept)	3.880	0.076	50.961	0.00000
bty_avg	0.067	0.016	4.090	0.00005

Let's confirm the p-value calculation:

$$\Pr(T \geq 4.09) + \Pr(T \leq -4.09) \quad \text{where } T \sim t_{461}$$

- Write out the code you would use to calculate the p-value.
- $2 * (1 - \text{pt}(4.09, \text{df} = 461)) = 0.00005$
- **Assuming the LINE conditions are met:** since our p-value 0.00005 is extremely small, we would reject H_0 at any reasonable significant level. Thus, the data provide convincing evidence that there is a linear relationship between instructor's beauty score and evaluation score.

Different H_A

term	estimate	std.error	statistic	p.value
(Intercept)	3.880	0.076	50.961	0.00000
bty_avg	0.067	0.016	4.090	0.00005

Write code for your p-value if your alternative was $H_A : \beta_1 > 0$. What would your conclusion be?

- $\Pr(T \geq 4.09) = 1 - \text{pt}(4.09, 461) = 0.000025$
- The data provide convincing evidence that there is a *positive* relationship between instructor's beauty score and evaluation score.

Write code for your p-value if your alternative was $H_A : \beta_1 < 0$. What would your conclusion be?

- $\Pr(T \leq 4.09) = \text{pt}(4.09, 461) = 0.9999745$
- The data do *not* provide convincing evidence that there is a *negative* relationship between instructor's beauty score and evaluation score.

Confidence intervals

term	estimate	std.error	statistic	p.value
(Intercept)	3.880	0.076	50.961	0.00000
bty_avg	0.067	0.016	4.090	0.00005

We can also construct confidence intervals using the output from `lm()`! Remember:

$$\text{CI} = \text{point est.} \pm \text{critical value} \times \widehat{\text{SE}} \quad (\text{or SE if we have it})$$

- Critical value also comes from t_{n-2} distribution
- Suppose we want a 95% confidence intervals for β_1 :
 - What code would you use to obtain critical value? Then set up your CI!
 - `qt(0.975, 461) = 1.97`

$$95\% \text{ CI} : 0.067 \pm 1.97 \times 0.016 = (0.035, 0.099)$$

Remarks

- Note: for β_1 , the null hypothesis is **always** of the form $H_0 : \beta_1 = 0$
- LINE conditions must be met for underlying mathematical and probability theory to hold here! If not met, interpret and perform inference with caution
- Here, the Independence and Normality conditions did not seem to be met
 - Take STAT 412 or other course to learn how to incorporate dependencies between observations!
- **So what can we say?**
 - The results suggested by our inference should be viewed as preliminary, and not conclusive
 - Further investigation is certainly warranted!
 - Checking LINE can be very subjective, but that's how real-world analysis will be!